

Principles of Cation– π Interactions for Engineering Mussel-Inspired Functional Materials

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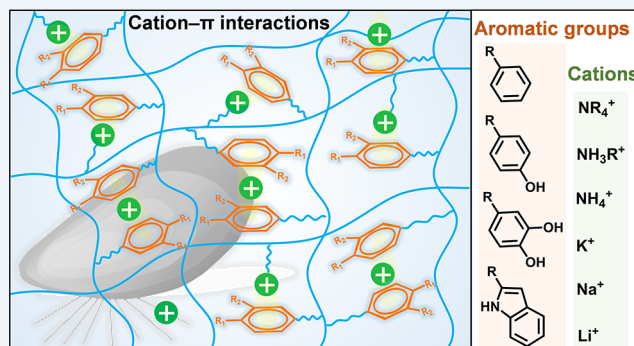
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CONSPECTUS: Supramolecular assembly is commonly driven by noncovalent interactions (e.g., hydrogen bonding, electrostatic, hydrophobic, and aromatic interactions) and plays a predominant role in multidisciplinary research areas ranging from materials design to molecular biology. Understanding these noncovalent interactions at the molecular level is important for studying and designing supramolecular assemblies in chemical and biological systems. Cation– π interactions, initially found through their influence on protein structure, are generally formed between electron-rich π systems and cations (mainly alkali, alkaline-earth metals, and ammonium). Cation– π interactions play an essential role in many biological systems and processes, such as potassium channels, nicotinic acetylcholine receptors, biomolecular recognition and assembly, and the stabilization and function of biomacromolecular structures. Early fundamental studies on cation– π interactions primarily focused on computational calculations, protein crystal structures, and gas- and solid-phase experiments. With the more recent development of spectroscopic and nanomechanical techniques, cation– π interactions can be characterized directly in aqueous media, offering opportunities for the rational manipulation and incorporation of cation– π interactions into the design of supramolecular assemblies. In 2012, we reported the essential role of cation– π interactions in the strong underwater adhesion of Asian green mussel foot proteins deficient in L-3,4-dihydroxyphenylalanine (DOPA) via direct molecular force measurements. In another study in 2013, we reported the experimental quantification and nanomechanics of cation– π interactions of various cations and π electron systems in aqueous solutions using a surface forces apparatus (SFA). Over the past decade, much progress has been achieved in probing cation– π interactions in aqueous solutions, their impact on the underwater adhesion and cohesion of different soft materials, and the fabrication of functional materials driven by cation– π interactions, including surface coatings, complex coacervates, and hydrogels. These studies have demonstrated cation– π interactions as an important driving force for engineering functional materials. Nevertheless, compared to other noncovalent interactions, cation– π interactions are relatively less investigated and underappreciated in governing the structure and function of supramolecular assemblies. Therefore, it is imperative to provide a detailed overview of recent advances in understanding of cation– π interactions for supramolecular assembly, and how these interactions can be used to direct supramolecular assembly for various applications (e.g., underwater adhesion). In this Account, we present very recent advances in probing and applying cation– π interactions for mussel-inspired supramolecular assemblies as well as their structural and functional characteristics. Particular attention is paid to experimental characterization techniques for quantifying cation– π interactions in aqueous solutions. Moreover, the parameters responsible for modulating the strengths of cation– π interactions are discussed. This Account provides useful insights into the design and engineering of smart materials based on cation– π interactions.



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E853.² The first instance of coacervation between two positively charged polyelectrolytes, which is triggered by short-range cation– π interactions by overcoming repulsive electrostatic interactions, is reported.

- Kim, S.; Yoo, H. Y.; Huang, J.; Lee, Y.; Park, S.; Park, Y.; Jin, S.; Jung, Y. M.; Zeng, H.; Hwang, D. S.; Jho, Y. Salt triggers the simple coacervation of an underwater adhesive when cations meet aromatic π electrons in seawater. *ACS Nano* **2017**, *11*, 6764–6772.³ This work reports a fluid–fluid separation phenomenon of a single cationic recombinant mussel foot protein without additional anionic polymers in seawater conditions, which is induced by cation– π interaction.
- Yan, B.; He, C.; Chen, S.; Xiang, L.; Gong, L.; Gu, Y.; Zeng, H. Nanoconfining cation– π interactions as a modular strategy to construct injectable self-healing hydrogel. *CCS Chem.* **2021**, *3*, 2903–2916.⁴ This work presents a facile and novel strategy to fabricate injectable self-healing synthetic hydrogel from the self-assembly of a thermoresponsive ABA triblock copolymer via cation– π interactions, which demonstrates a new strategy for developing self-healing materials by incorporating cation– π interactions with various engineering and bioengineering applications.

1. INTRODUCTION

Noncovalent interactions are ubiquitous in nature and govern numerous important biological processes, such as the self-assembly of DNA double helix structures, the folding of proteins, and different forms of molecular recognition. Various noncovalent interactions, including hydrogen bonding, hydrophobic, van der Waals, and aromatic interactions, also play a critical role in many areas of materials science, influencing the formation of supramolecular complexes and molecular crystals, and mechanical properties of adhesive or hydrogel materials.⁵ Noncovalent interactions have been widely recognized for their contributions to the design and assembly of natural and synthetic materials, thus understanding these interactions at the nanoscale or molecular level is crucial for applying them in fabricating functional materials. Compared to the other noncovalent interactions that have been studied extensively, cation– π interactions were only discovered in the 1980s. Cation– π interactions, formed between various positively charged cations and the negatively charged electron clouds of π systems, play an essential role in many biological systems, such as potassium channels, nicotinic acetylcholine receptors, and the stabilization and function of many protein structures.^{6,7} In addition, cation– π interactions with low desolvation energies contribute significantly to the strong underwater adhesion of marine organisms, such as mussels.^{8,9} These examples demonstrate the significance and importance of cation– π interactions in biological systems and provide a useful driving force for fabricating biomimetic synthetic materials.

Over the last few decades, significant progress has been achieved in probing cation– π interactions in various material systems. Dougherty and co-workers pioneered theoretical simulations of cation– π interactions in molecular neurobiology, which significantly advanced the fundamental understanding of the origins and characteristics of cation– π interactions.⁶ A comprehensive review by Sastry and co-workers focused on the parameters responsible for modulating cation– π interactions along with their structural and functional significance.⁷ It is

worth noting that most previous studies on cation– π interactions mainly focused on theoretical research, and some recent reviews discussed the role of cation– π interactions in controlling the regio- and stereoselectivities in organic synthesis,¹⁰ governing the conformation and arrangement of small molecules in crystals,¹¹ maintaining the structure and function of mussel foot proteins,⁹ and dominating the structural stability and selectivity of graphene-based membranes.¹² Experimental characterization and force measurements of cation– π interactions have provided useful insights into the fundamental understanding of the supramolecular assembly of functional materials (e.g., surface coatings and hydrogels) driven by such interactions. These important aspects have not yet been fully covered by previous reviews, and thus it is important to provide an overview of the experimental quantification and the underlying design principles of cation– π interactions for engineering functional materials.

Given the growing interest in cation– π interactions, this Account highlights their importance in materials engineering. Moreover, we share our perspectives gained from a decade of research on cation– π interactions, with a focus on cation– π binding pairs formed by electron-rich π systems (e.g., benzene, phenol, catechol, and indole residues) and cations, such as alkali metals (e.g., K^+ and Na^+) and nitrogenous cations (e.g., $-NH_3^+$ and $-NMe_3^+$). Specifically, this Account includes the following aspects: (i) experimental advances on mussel-inspired cation– π interactions for engineering functional materials; (ii) characterization techniques for cation– π interactions and parameters that influence the self-assembly process of cation– π mediated materials; (iii) various supramolecular assemblies (e.g., films, hydrogels, and complex coacervates) formed and driven by cation– π interactions (Figure 1).

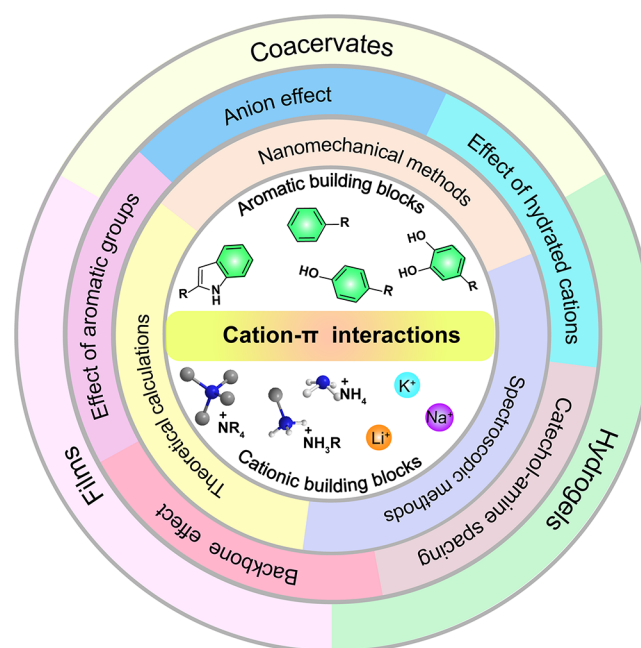


Figure 1. Overview of building blocks, parameters, and characterization methods for engineering supramolecular assemblies based on cation– π interactions.

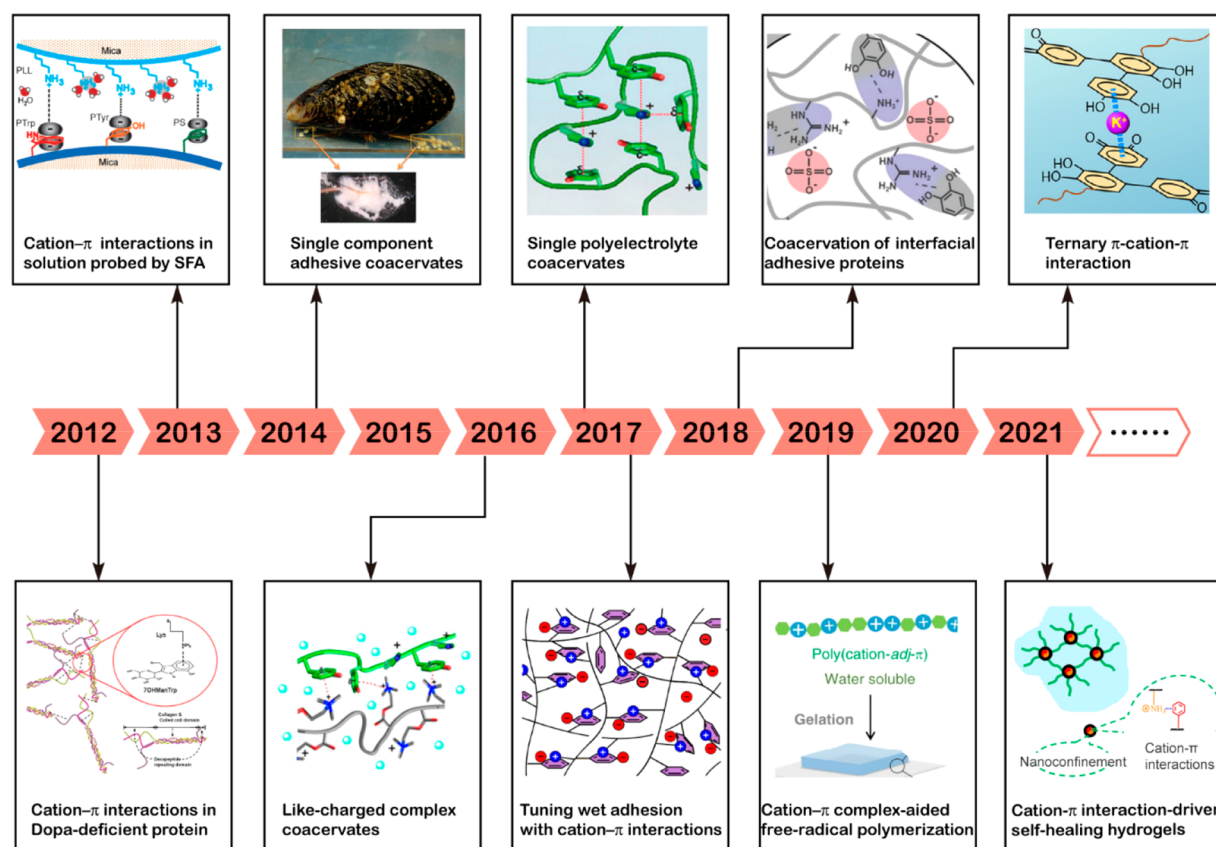


Figure 2. Timeline illustrating the advances in understanding mussel-inspired cation- π interactions starting from 2012. Adapted with permission from the following references: ref 1, Copyright 2013 Wiley-VCH; ref 2, Copyright 2016 National Academy of Sciences; ref 3, Copyright 2017 American Chemical Society; ref 4, Copyright 2021 Chinese Chemical Society; ref 8, Copyright 2012 Royal Society of Chemistry; ref 28, Copyright 2014 Elsevier; ref 29, Copyright 2017 Springer Nature; ref 30, Copyright 2019 Springer Nature; ref 31, Copyright 2020 Elsevier; ref 32, Copyright 2018 Wiley-VCH.

2. ADVANCES IN CATION- π INTERACTIONS

It has been more than 40 years since the first report of cation- π interactions in 1981, in which it was found that the positively charged K^+ could bind to benzene in preference to negative ions in the gas phase.¹³ In the early studies, Dougherty and co-workers made significant progress on revealing the relevance of cation- π interactions to molecular recognition in various host-guest model systems.^{5,14} For example, cyclophanes can pull quaternary ammonium and immonium compounds out of water and into their well-defined hydrophobic cavities.^{15,16} These results demonstrated that cation- π interactions, instead of hydrophobic effects, drove the supramolecular assembly via binding organic molecules within the hydrophobic cavities of cyclophane macrocycles, with great implications for molecular recognition in biology. Direct and circumstantial evidence indicates that phenylalanine (Phe), tyrosine (Tyr), and tryptophan (Trp) can provide potent binding sites in proteins for cations to form cation- π interactions, which occur in a wide range of biological processes.¹⁷ For example, cationic acetylcholine, an important neurotransmitter, shows a strong affinity for proteins with conserved aromatic residues, such as acetylcholine esterases.^{18,19} Cation- π interactions also contribute to the ion selectivity in potassium channels, transport rates of ammonia transporters, and high affinity for nicotine in the brain.^{20–22}

A remarkable feature of marine organisms is their moisture-resistant adhesion to various surfaces. Most mussel foot proteins are rich in L-3,4-dihydroxyphenylalanine (DOPA) as well as lysine (Lys), which can interact synergistically to promote wet

adhesion by surface salt displacement.^{23,24} Apart from enhancing adhesion, cationic residues and their close proximity to DOPA result in short-range cation- π interactions, which contribute to the strong cohesive and mechanical properties as well as the stable structures of the mussel byssal threads in marine environments (e.g., high salinity and alkalinity, wave erosion).^{25,26} Even though the foot protein of Asian green mussels lacks DOPA residues, the backbones of the proteins are still rich in Trp derivatives and can initiate effective cation- π interactions between the indole rings of Trp and Lys to induce underwater adhesion.⁸ Through molecular force measurements, we demonstrated the importance of cation- π interactions in the adhesion and cohesion properties of mussel foot proteins.^{8,27}

Cation- π interactions in various biological and chemical systems have attracted great interest, especially in the area of mussel-inspired phenolic chemistry. Over the past decade, significant advances have been achieved in understanding, controlling, and designing cation- π interactions in materials science (Figure 2).

3. CHARACTERIZATION AND MODULATING PARAMETERS OF CATION- π INTERACTIONS

Computational studies have provided significant insights into understanding the binding nature of cation- π interactions. Dougherty and co-workers used *ab initio* molecular orbital calculations and AM1 methods to investigate 17 different cation- π complexes, which demonstrated that the binding energies were strongly influenced by electrostatic forces.³³ It was

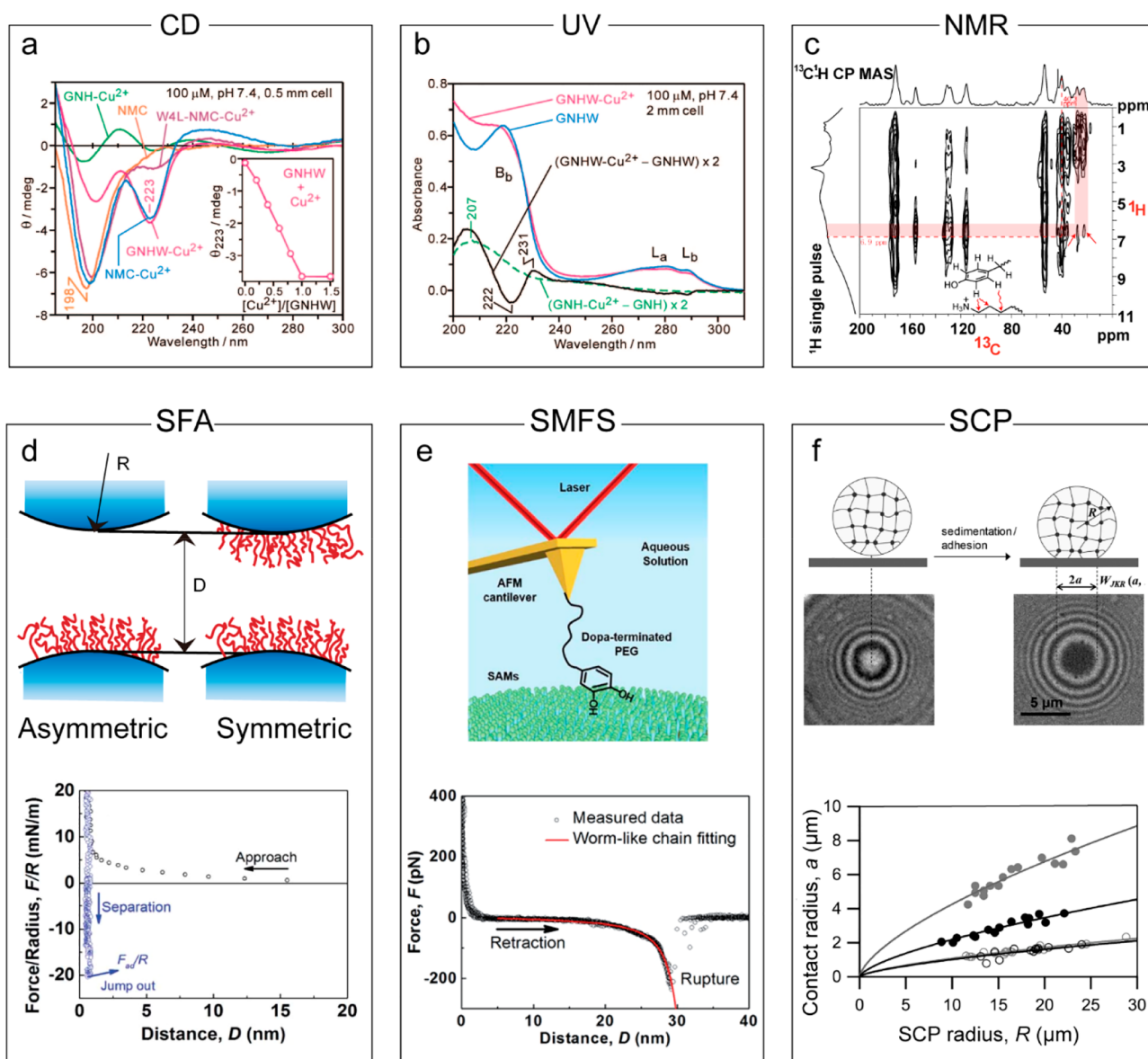


Figure 3. Characterizations of cation- π interactions using different techniques. (a, b) CD and UV spectra for cation- π interactions between Cu^{2+} and Trp. Adapted with permission from ref 38. Copyright 2008 American Chemical Society. (c) Solid-state 2D ^{13}C - ^1H HETCOR MAS NMR spectrum of a Tyr-rich polypeptide. Adapted with permission from ref 29. Copyright 2017 Springer Nature. (d) SFA scheme and force-distance curves between mica surfaces in the presence of poly[(2-methacryloyloxyethyl phosphorylcholine)-*co*-(dopamine methacrylamide)]. (e) SMFS scheme and force-distance retraction curves for a single DOPA molecule interaction with 2-methacryloyloxyethyl phosphorylcholine. Panels d and e adapted with permission from ref 45. Copyright 2019 The Royal Society of Chemistry. (f) SCP scheme and contact radius-SCP radius curves of PEG-crotonic acid SCPs adhering on poly(ethylene imine) surfaces. Adapted with permission from ref 46. Copyright 2017 Wiley-VCH.

suggested that cation- π interactions are essentially electrostatic attractions.^{17,34} To date, the electrostatic model is still one of the popular tools to predict the binding ability of new cation- π systems. Considering the complex physics and multiple influencing factors of cation- π interactions, various methods, such as density functional theory, second-order Møller-Plesset perturbation theory, and Hartree-Fock theory, have also been applied to investigate cation- π interaction energies.^{35–37} According to computer simulations, the binding energies of model systems for different cations follow the trend of $\text{Mg}^{2+} > \text{Ca}^{2+} > \text{Li}^+ > \text{Na}^+ > \text{K}^+ \approx \text{NH}_4^+ > \text{NMe}_4^+$ in the gas phase.³⁷ Moreover, the theoretical strength of cation- π interactions is two times larger than that of salt bridges in aqueous media, highlighting their importance for protein engineering.³⁵

In the gas phase, high-pressure mass spectrometry and ion cyclotron resonance are commonly used to characterize the binding of cations to aromatic molecules.^{5,9} Alternatively, X-ray techniques are usually employed to examine cation- π interactions in protein crystal structures.⁹ However, when compared to that in the gas phase, the cation- π interaction strengths are generally much weaker in aqueous media due to the solvation (i.e., hydration) of the cations. Various spectroscopic techniques have been applied to characterize cation- π interactions in aqueous solutions. Circular dichroism (CD) spectroscopy has been used to demonstrate the negative band at 223 nm attributed to cation- π interactions between Cu^{2+} and Trp (Figure 3a).³⁸ Similarly, Cu^{2+} -Trp interactions result in a negative band at 222 nm and a weak positive band at 231 nm in the ultraviolet (UV) difference spectra (Figure 3b).³⁸

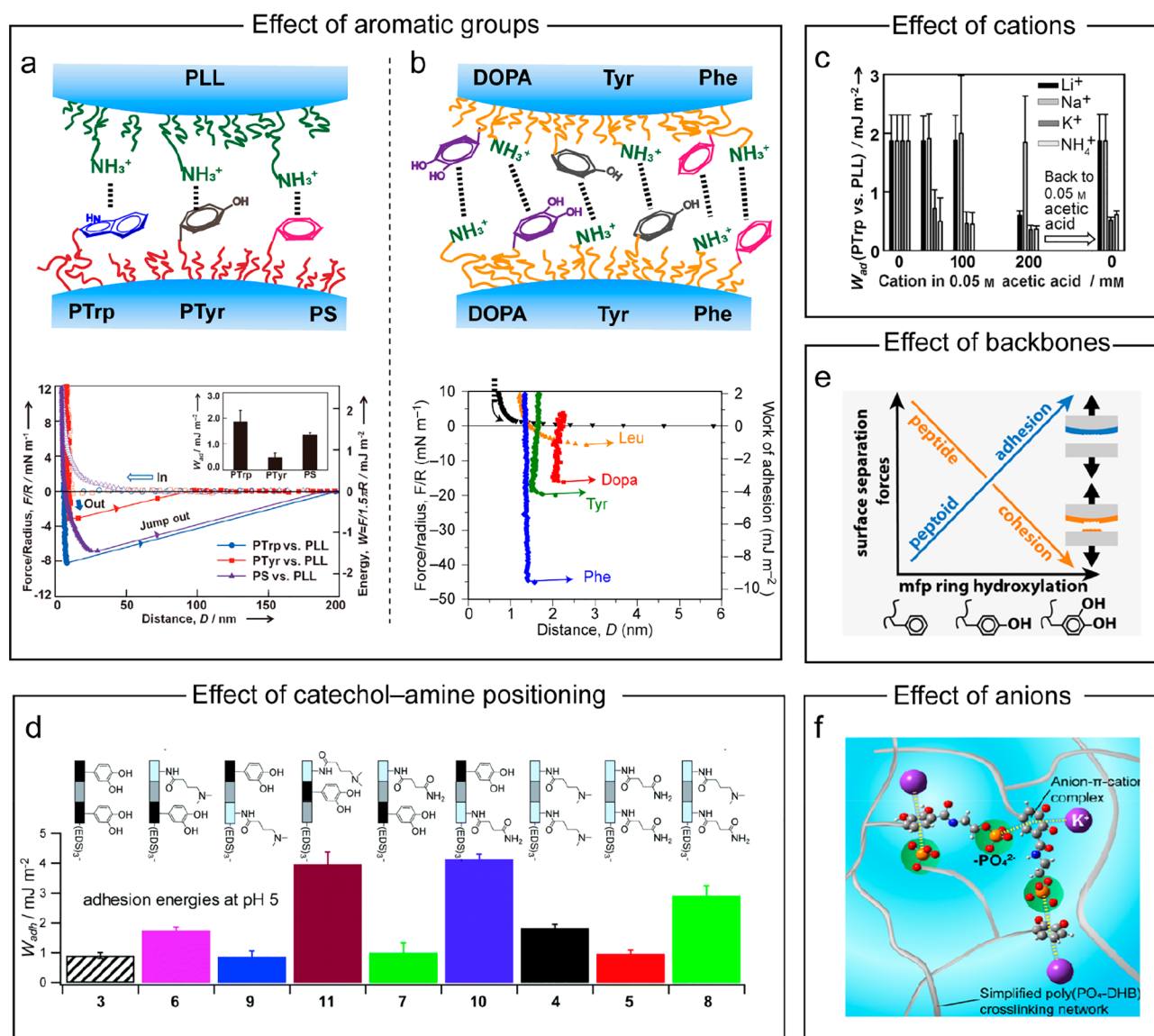


Figure 4. Influence of different parameters on the strength of cation- π interactions. (a) Schematic illustration and force-distance curves of the nanomechanics of PLL-aromatic polymer (PS, PTyr, or PTrp). (b) Schematic illustration and force-distance curves of the cohesion of peptides containing Lys and aromatic (Phe, Tyr, or DOPA) side chains. Effects of (c) hydrated cations, (d) catechol-amine positioning, and (e) anions on the strength of cation- π interactions. Effects of (c) hydrated cations, (d) catechol-amine positioning, and (e) anions on the strength of cation- π interactions. Panels a and c adapted with permission from ref 1. Copyright 2013 Wiley-VCH. Panel b adapted with permission from ref 29. Copyright 2017 Springer Nature. Panel d adapted with permission from ref 47. Copyright 2020 Royal Society of Chemistry. Panel e adapted with permission from ref 52. Copyright 2020 American Chemical Society. Panel f adapted with permission from ref 43. Copyright 2020 American Chemical Society.

UV resonance Raman spectroscopy has been applied to characterize the formation of coacervates, where the enhancement in relative intensity of the bands at 834, 856, 1185, 1267, and 1620 cm⁻¹ can be attributed to the cation- π interactions between Tyr and Lys in recombinant mussel foot protein, rmfp-1, in the presence of NaCl.³ Additionally, nuclear magnetic resonance (NMR) spectroscopy has been extensively employed to characterize cation- π complexes, where high field shifts of the proton peaks of aromatic rings (¹H NMR spectra) occurs in cation- π systems instead of the pairs of neutral groups (e.g., dimethylamino residues) and aromatic rings, indicating the molecular influence of cations on cation- π interactions.³⁰ The molecular proximities and orientations of Lys and aromatic Tyr in peptides can be identified in solid-state 2D ¹³C-¹H heteronuclear correlation experiments, which also confirm the strong cation- π interactions in peptides (Figure 3c).²⁹

Although the existence of cation- π interactions in aqueous solutions was demonstrated by spectroscopic measurements, it remained a challenging issue to experimentally quantify and compare the interaction strengths of various cation- π systems, as other noncovalent interactions generally could coexist, thus causing difficulty to isolate the contributions of the cation- π interactions. A surface forces apparatus (SFA) is a nano-mechanical technique that has been widely applied to quantify the intermolecular and surface forces (e.g., van der Waals, electrical double layer forces, hydrophobic interaction) as a function of the surface separation in different material systems.^{39,40} Two curved surfaces in crossed-cylinder configuration are commonly employed in SFA measurements, and their interaction forces can be correlated to the adhesion energy (W_{ad} for the asymmetric configuration) or cohesion energy (W_{co} for the symmetric configuration) per unit area between two

parallel planes of the same material(s) (Figure 3d).^{27,41} In 2013, the cation– π interactions between various cations and aromatic groups in aqueous solutions were experimentally quantified using SFA.¹ Recently, the SFA technique was further employed to characterize the contribution of primary amine groups to adhesion of phenolic coatings, and to demonstrate the existence of a ternary π –cation– π interaction in poly(catechol) systems and the cooperativity between cation– π interactions and anion– π interactions.^{31,42,43} Single-molecule force spectroscopy (SMFS) based on atomic force microscopy (AFM) is a complementary technique to SFA and has been widely used to quantify the forces on various molecules, for example, hydrogen bonding (e.g., DOPA–silica interactions) and coordination interactions (e.g., catechol–Fe³⁺ coordination).⁴⁴ The zwitterion–DOPA bond dissociation energy was measured to be $\sim 19.4k_B T$ using SMFS, which was attributed to the synergy between cation– π interactions and hydrogen bonding (Figure 3e).⁴⁵ However, the bond dissociation energy was halved when the cation– π interactions were shielded.⁴⁵ Soft colloidal probes (SCPs) (e.g., poly(ethylene glycol) (PEG) hydrogel particles) can undergo mechanical deformation when adhering to a substrate surface, which is related to their adhesion energies (Figure 3f).⁴⁶ SCP adhesion tests were used to characterize the adhesion strength of sequence-defined peptides against glass surfaces, which demonstrated that the distance between catechol and amine residues on polymer chains had a significant effect on adhesion strength.⁴⁷ These mechanical force techniques provide new avenues to understand the nano-mechanics of cation– π interactions in aqueous media for biological and materials assembly.

The parameters for modulating the strength of cation– π interactions are discussed in this section based on experimental evidence provided by mechanical techniques, such as SFA, SMFS, and SCP. The effect of aromatic groups on the adhesion strength in aqueous solutions was examined by SFA in an asymmetric experimental configuration (Figure 3d).^{1,9} Specifically, surfaces of mica were coated either with poly(L-lysine) (PLL) containing typical cations ($-\text{RNH}_3^+$) or with aromatic polymers, including poly(L-tryptophan) (PTrp), poly(L-tyrosine) (PTyr), or polystyrene (PS). These measurements allowed the determination of how the polymer architecture influenced the adhesion energy, where W_{ad} was found to be ~ 1.8 , ~ 1.3 , and ~ 0.5 mJ/m² for PLL–PTrp, PLL–PS, and PLL–PTyr, respectively (Figure 4a).^{1,9} The SFA measurements demonstrated that the adhesion strength between PLL and different aromatic polymers followed the order PTrp > PS > PTyr, and the relative binding energies correlated well with the electrostatic potential energy of the aromatic residues. A series of lysine- and aromatic residue-rich peptides were then designed and used to examine the effect of hydroxyl substituents (e.g., Phe, Tyr, and DOPA) on cation– π binding cohesive energies via SFA with a symmetric configuration. Resulting W_{co} of Phe-rich peptides, Tyr-rich peptides, and DOPA-rich peptides were ~ 10 , ~ 4.0 , and ~ 3.6 mJ/m², respectively (Figure 4b).²⁹ Interestingly, the cohesion of Phe-rich peptides was twice that of either Tyr- or DOPA-rich peptides, which could be due to the high configurational entropy within Phe-rich peptides.²⁹ These results demonstrate that the cohesion strength is dependent on the presence of aromatic-ring hydroxylation.

Generally, cation– π interactions are also affected by the presence of other cations, where the binding force can be undermined due to their interfering effect. We investigated the influence of monovalent cations on the adhesive interactions in

model systems (e.g., PLL–PTrp and poly(catechol)–poly(catechol)),^{1,31} and the results showed that the affinity of cations with aromatic groups followed the order of $\text{NH}_4^+ \approx \text{K}^+ > \text{Na}^+ > \text{Li}^+$ (Figure 4c). It should be noted that such an order was not the same as that found in the gas phase, due to the hydration effect of the cations in aqueous media.^{17,20} Such molecular force measurements provide useful insights into various biological phenomena, such as potassium channel blockers possessing $-\text{NR}_4^+$ and Lys promoting the underwater adhesion of mussel foot proteins by surface salt displacement.^{1,17,24}

Controlling the positioning of catechol and amine groups in a polymer chain is an effective method to modulate adhesive performance.^{47–51} SCP adhesion assays of sequence-defined polymers demonstrated that the adhesion energy of a polymer with adjacent catechol and amine residues was significantly higher than that of polymers with an additional spacer between the catechol and amine groups (Figure 4d).⁴⁷ Moreover, terminal amides can displace the surface hydration layer on substrates and provide a synergistic effect with catechols for underwater adhesion, which is attributed to the ionic resonance structure of primary amides.⁴⁷ Messersmith and co-workers performed macroscopic adhesion tests, which demonstrated that the close proximity of cation–aromatic pairs produced cooperative effects in improving macroscale cohesive strength.⁴⁸ Overall, the cooperative effects between phenol and amine groups provide new avenues for the rational design of functional materials.

The polymer backbone structure of π systems is another factor that influences cation– π interactions. SFA measurements of mussel-mimetic adhesive peptides and analogous peptoids with different numbers of hydroxyl groups on benzene rings demonstrated the significant impact of backbone structure on adhesive and cohesive strength, where more hydroxyl groups resulted in higher adhesion for peptoids but less cohesion for peptides (Figure 4e). For example, the adhesion of Phe-rich peptoids is 5-times weaker than that of Phe-rich peptides, while DOPA-rich peptoids have at least 2-fold stronger adhesion compared to DOPA-rich peptides.⁵² In addition, anions can also influence the strength of cation– π interactions. Excessive cations can induce the decrease of adhesion and coating thickness of phenolic coatings;³¹ however the presence of anionic phosphate ester groups can stabilize cation– π binding pairs through anion– π interactions (Figure 4f).⁴³ As a result, anion– π –cation interactions are barely affected by excessive cations (i.e., K^+), which indicates that anion– π –cation complexes are able to tolerate high concentrations of cations. The effect of anions on cation– π interactions demonstrates a promising way to mediate the adhesion and deposition of phenolic coatings.

4. CATION– π INTERACTIONS FOR MATERIALS ENGINEERING

Mussel-inspired chemistry has been well recognized as a promising strategy to design and exploit advanced materials for applications in biomedical, energy, and environmental fields.^{53,54} Inspired by the multifunctional roles of cation– π interactions in mussel adhesive proteins, a wide range of assemblies (e.g., coacervates, hydrogels, and films) based on cation– π interactions have been developed. The persistent adhesive performance allows widespread application of cation– π -based materials in underwater adhesion and tissue engineering (e.g., wound sealing, hemostasis).^{55,56} In addition,

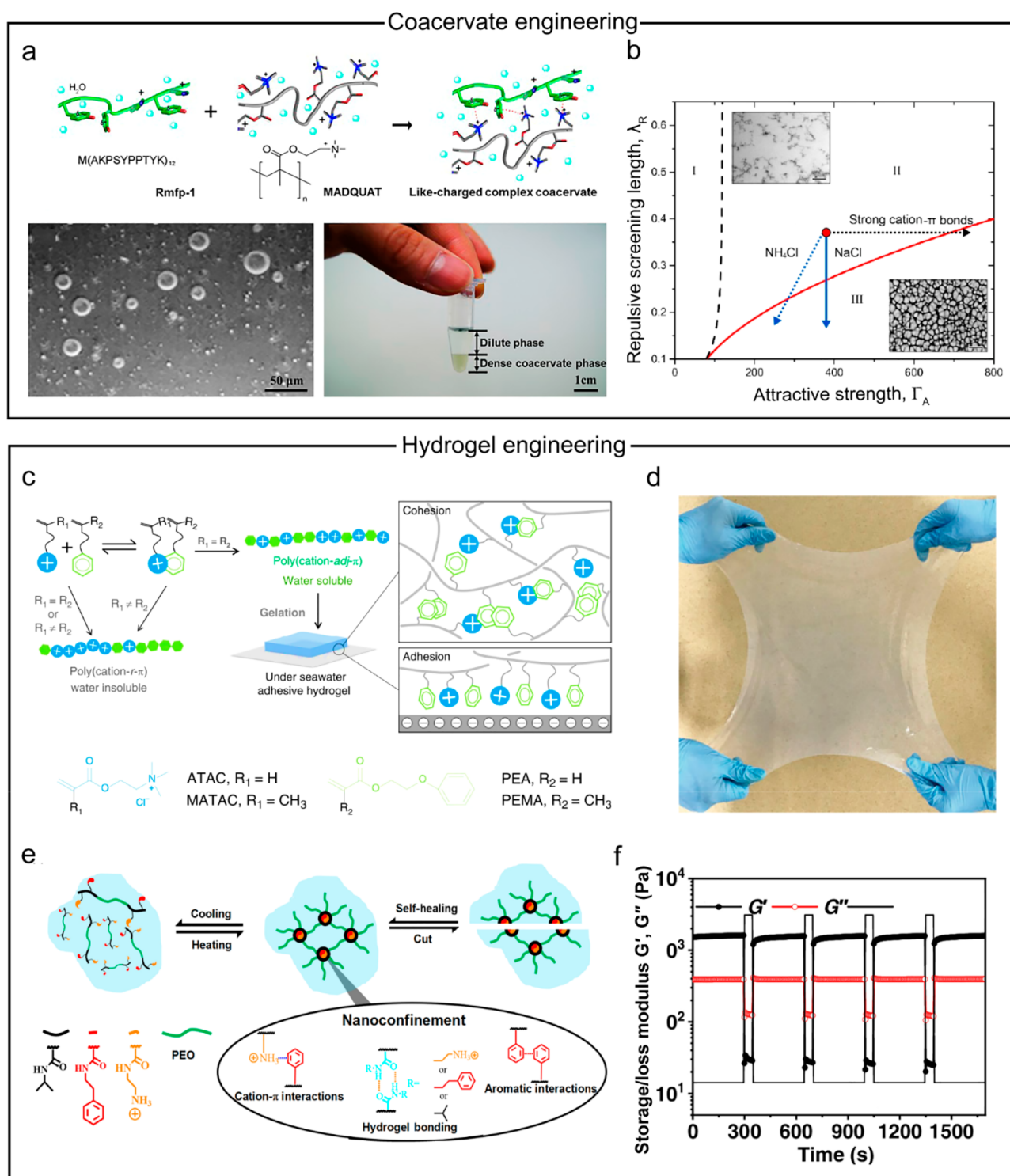


Figure 5. (a) Formation of like-charged coacervates. Adapted with permission from ref 2. Copyright 2016 National Academy of Sciences. (b) Phase diagram of *rmfp-1* for the formation of (I) liquid-like, (II) finite complex, and (III) infinite complex phases. Adapted with permission from ref 3. Copyright 2017 American Chemical Society. (c) Synthesis of poly(cation-adj- π) copolymers by using adjacent cationic and aromatic monomers based on cation- π interactions. (d) Photograph of poly(cation-adj- π) hydrogels. Adapted with permission from ref 30. Copyright 2019 Springer Nature. (e) Scheme showing the assembly of thermoresponsive and self-healing hydrogels via cation- π interactions. (f) Dynamic strain cyclic test of the self-healing hydrogels at 37 °C. Adapted with permission from ref 4. Copyright 2021 Chinese Chemical Society.

cation- π -driven 2D self-assemblies were used as an effective carrier of Pd-nanoparticles for catalytic applications.⁵⁷

4.1. Coacervates

Coacervation refers to a liquid-liquid phase separation phenomenon, and conventional coacervates with low fluid

viscosity and low interfacial tensions are composed of oppositely charged polyelectrolytes and formed through electrostatic interactions. Although most mussel foot proteins are cationic polyelectrolytes, they can solidify to form dense coacervates upon contact with seawater due to cation- π interactions

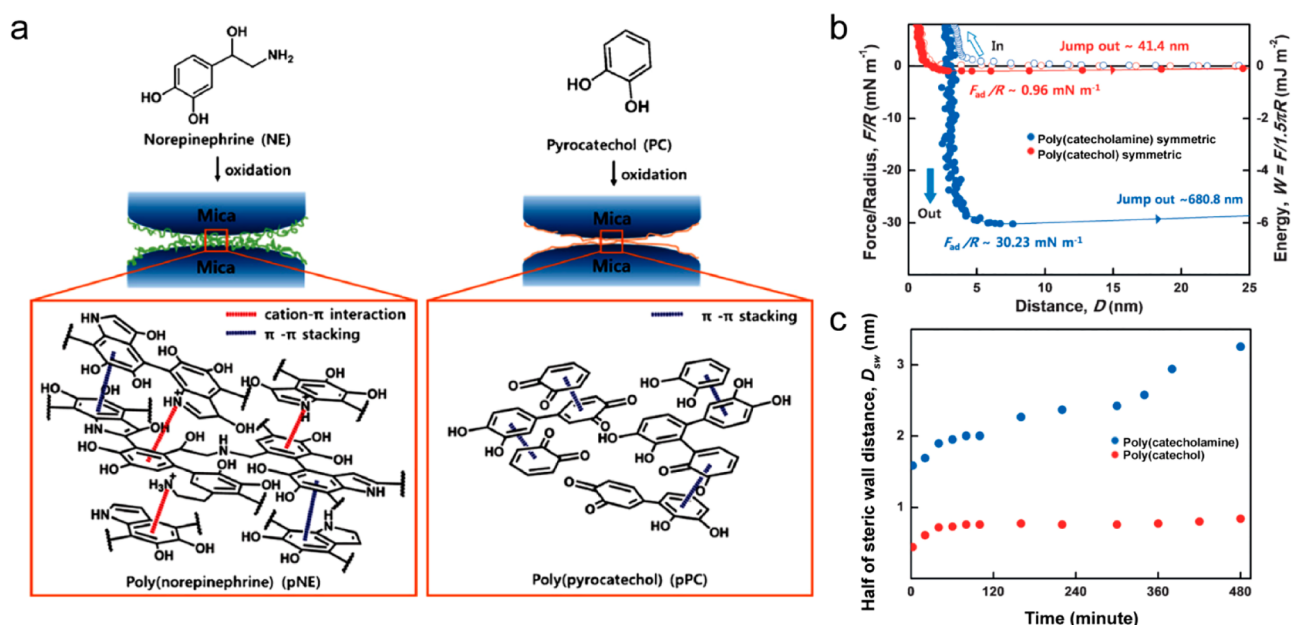


Figure 6. (a) Nanomechanical measurements of poly(catecholamine) and poly(catechol) films. (b) Representative force–distance curves and corresponding interaction energy between two opposing poly(catecholamine) or poly(catechol) layers. (c) Time-dependent film thickness of poly(catecholamine) and poly(catechol) films deposited on mica. Adapted with permission from ref 42. Copyright 2016 Wiley-VCH.

between cationic residues and aromatic groups on the adhesive proteins.^{3,28} This like-charged coacervation or self-coacervation phenomenon is vital for the initial mussel adhesion process.³ The synthesis of like-charged coacervates can therefore be accomplished by mixing phenolic rmfp-1 and cationic polymers containing trimethylammonium in aqueous media, where coacervation is triggered by strong cation- π interactions overcoming electrostatic repulsion (Figure 5a).² Interestingly, the interfacial energy of the like-charged coacervate is <0.57 mJ/m², which is as low as that of conventional coacervates composed of oppositely charged polyelectrolytes, and therefore the like-charged coacervates could be readily spread on various substrates and used for coating nanoparticles to prevent nanoparticle aggregation.²

Self-coacervates were obtained when the zwitterionic mussel foot protein-3S (Mfp-3S) was used via increasing both pH and ionic strength, without the need for anionic proteins or synthetic molecules.²⁸ The Mfp-3S coacervates have low interfacial energies (≤ 1 mJ/m²) and a high adsorption capacity on dental and orthopedic surfaces. Besides zwitterionic proteins, positively charged mussel proteins, such as foot protein type-3 fast variant and rmfp-1, can also form self-coacervates in saline media.^{3,32} The phase diagram of the rmfp-1 model system demonstrated that the transformation of a finite complex of rmfp-1 to the coacervate phase can be triggered by an increase in cation- π bond strength or a decrease in long-distance repulsive electrostatic interactions (Figure 5b).³ These findings are helpful to understand the molecular mechanisms of mussel adhesion and demonstrate that the formation of like-charged coacervates or self-coacervates driven by cation- π interactions is influenced by a range of variables. Finally, the low fluid viscosity and low interfacial tension endow coacervates with promise for underwater coatings as well as the encapsulation and dispersion of particles and cells.

4.2. Hydrogels

Mussel-inspired adhesive hydrogels have been extensively investigated, while the fabrication of hydrogels driven by

cation- π interactions is rarely reported, likely because cation- π interactions are a short-range attractive force, which can hinder three-dimensional network formation and stabilization. Therefore, cation- π interactions are typically combined with other interactions to form hydrogels with improved strength. For instance, mechanically strong hydrogel fibers (breaking strength of 160 MPa, Young's modulus of 17 GPa, and toughness of 99 MJ/m³) were fabricated due to the synergy of multiple noncovalent interactions, including metal coordination and cation- π interactions.⁵⁸

Recently, Gong and co-workers synthesized poly(cation-adj- π) copolymers using adjacent cationic and aromatic monomers preassembled through cation- π interactions (Figure 5c).³⁰ Due to the cation- π interactions between adjacent aromatic rings and cations, as well as intra- and intermolecular π - π interactions, poly(cation-adj- π) materials possess strong cohesive interactions to induce the formation of hydrogels in saline media (Figure 5d). In the bulk hydrogel, cation- π and π - π interactions provide dynamic bonds to enhance both the toughness and the cohesion strength. Cooperative interactions between aromatic residues and adjacent cations in poly(cation-adj- π) can also screen electrostatic interactions and promote robust and reversible underwater adhesion.⁵⁹ The influence of small cations (i.e., Na⁺) in the surrounding environment was examined by measuring the Donnan potential values,⁶⁰ and it was seen that the phenyl groups on the polymer chains can bind with Na⁺ through cation- π interactions despite the existence of adjacent macrocations, resulting in weaker poly(cation-adj- π) hydrogels.^{30,60} The competition between polymeric cations and the small cations and aromatic polymers can help us to better understand the association between biomolecules and metal ions in various biological systems.

To avoid the use of organic solvents and additional solvent exchange steps for constructing biocompatible hydrogels based on cation- π interactions as discussed above, we reported a facile strategy to fabricate injectable thermosensitive and self-healing hydrogels via assembly of polymer micelles based on cation- π

interactions (Figure 5e).⁴ The hydrogel demonstrated a self-healing efficacy of 96.5% (Figure 5f) and was significantly superior compared to hydrogels not involving cation– π interactions.⁴ The study provides a new approach for the engineering of responsive hydrogels by incorporating cation– π interactions.

4.3. Films

Mussel-inspired phenolic chemistry has been widely utilized as a versatile strategy to assemble films on various substrates.^{53,61} The contribution of primary amine groups to catechol-based coatings was investigated by utilizing catecholamine and pyrocatechol as model systems (Figure 6a).⁴² For example, the adhesion of a poly(catecholamine) coating was ~ 30 mN/m, which was 30 times higher than that of a poly(catechol) coating (Figure 6b). Moreover, poly(catecholamine) resulted in a denser film than poly(catechol) (Figure 6c). X-ray photoelectron spectroscopy (XPS) measurements demonstrated the existence of quaternary amines in the poly(catecholamine) coatings but not in the poly(catechol) coatings, where these can contribute to cation– π interactions to improve the adhesive strength.⁴² In addition, the adhesion of poly(catecholamine) films decreased with increasing K^+ concentrations, which suggested the competition of K^+ with the cation– π pairs. On the other hand, the adhesion strength and film thickness of poly(catechol) coatings increased from ~ 0.68 mN/m and ~ 1.5 nm to ~ 23.75 mN/m and ~ 7.5 nm, respectively, when the K^+ concentration increased from 0 to 250 mM. However, the adhesion strength and film thickness then decreased to ~ 5.23 mN/m and ~ 2.7 nm, respectively, when the K^+ concentration further increased from 250 to 600 mM.³¹ Appropriate K^+ concentration can therefore facilitate the self-assembly of poly(catechol) films via formation of ternary π –cation– π complexes, while excess cations could result in the transformation to binary cation– π complexes, as well as breaking bridging and undermining assembly. Specifically, the binding strength of poly(catechol) with various cation species follows the order of $NMe_4^+ > K^+ > Na^+ > Li^+$.

Cation– π interactions exist in polydopamine (PDA) films, which generally can be disassembled in alkaline conditions (i.e., pH 12) due to the deprotonation of the amine groups.⁶² More interestingly, the disassembled PDA films can be recovered by adding K^+ , which compensates for the loss of the cationic amine species in PDA. Moreover, the thickness, roughness, and water contact angles of PDA films are affected by the K^+ concentration. In addition to K^+ , NMe_4^+ also can reverse the delamination of PDA films, leading to the formation of material-independent superhydrophilic coatings.⁶² Similarly, robust coatings in aqueous media with high salinity and mild alkalinity can be formed by integrating primary amines (e.g., *o*-phenylenediamine, tetraethylenepentamine, and polyethylenimine) into PDA assemblies.²⁵ These results demonstrate the vital role of cations for cation– π interactions in film fabrication and highlight how the concentration and type of cations can modulate the assembly and subsequent properties of phenolic films.

5. SUMMARY AND OUTLOOK

In summary, this Account focuses recent advances on and fundamental design principles of cation– π interactions for the engineering of mussel-inspired functional materials (e.g., coacervates, hydrogels, and films) and presents our recent efforts on probing the nanomechanics of cation– π interactions

and fabricating a range of materials. Although cation– π interactions have been studied for the past few decades, the supramolecular assembly of advanced materials based on cation– π interactions has only been reported more recently, especially when compared to material assemblies driven by other noncovalent interactions. Thus, it is necessary to develop other functional materials beyond mussel-inspired chemistry to expand the potential of cation– π interactions. For example, cation– π interactions are common in graphene-based materials. The interplay between cations and the delocalized polarizable π electrons of graphene can alter the intrinsic characteristics of graphene, such as structural stability and interlayer spacing, which significantly affect the performance of graphene-based devices.^{12,63} We herein highlight several future directions for possibly further advancing this field. First, expanding the choice of π -electron donors is of practical relevance for studying the structural and energetic characteristics of cation– π interactions, as most π systems reported to date are limited to several candidates (e.g., DOPA, phenol, and benzene). For example, plant-inspired pyrogallol derivatives containing molecules with three hydroxyl substituents on an aromatic ring (e.g., gallic acid), could potentially provide binding sites for cations. Macromolecules or peptides composed of more than one type of π -electron donors, such as tannic acid or epigallocatechin gallate, also offer a wide range of possibilities to probe the innovative properties of cation– π interactions. Second, the incorporation of nanomechanical techniques, such as SFA and AFM, with spectroscopy characterization methods, such as X-ray scattering, Raman spectroscopy, or fluorescence microscopy, will enable the comprehensive characterization of cation– π interactions in more complex systems and environments. Third, the exploration of diverse supramolecular assemblies (e.g., metal–organic frameworks, superstructures) dominated by cation– π interactions is promising for providing new avenues for emergent applications. Overall, this Account provides an overview, useful insights, and perspectives on the fundamental understanding and assembly mechanisms of cation– π interactions toward engineering advanced functional materials.

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Notes

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